

BASIC ELECTRONICS

(24B11EC111)

Unit-1

Lecture-3

(Voltage Divider Rule)

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Voltage Divider Rule

Definition

- A **voltage divider** is a basic circuit, **used to generate a part of its input voltage like an output**. This circuit can be designed **with two resistors** otherwise **any passive components** along with a voltage source.
- The **resistors** in the circuit can be connected **in series** whereas a **voltage source is connected across these resistors**. This circuit is also called a **potential divider**.
- The input voltage can be transmitted between the two resistors in the circuit so that the division of voltage takes place.

Voltage Divider Calculation for two series resistor

- The connection of resistors in Figure 1 is said to be a series connection because all the elements carry the same current.
- To calculate the voltage across each resistance, we need to find the current in the loop ie. i_1
- To determine i_1 , we use KVL around the loop to obtain

$$V = V_1 + V_2$$

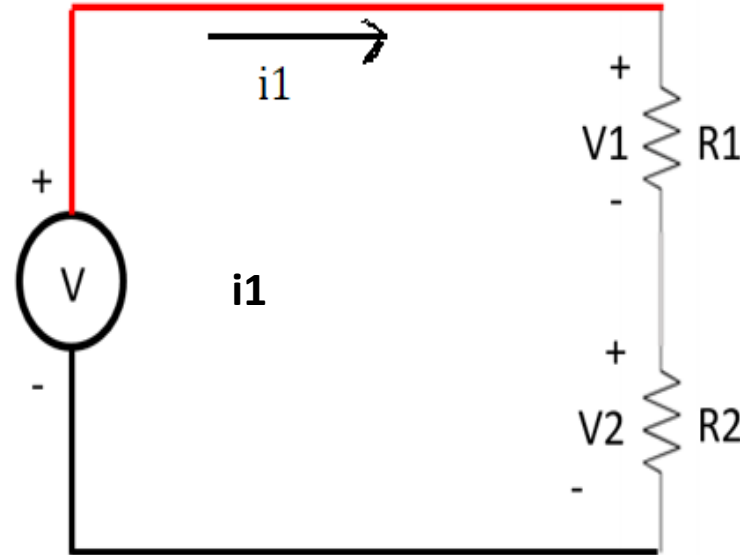


Figure1

Contd..

- Solving for i_1 , we have

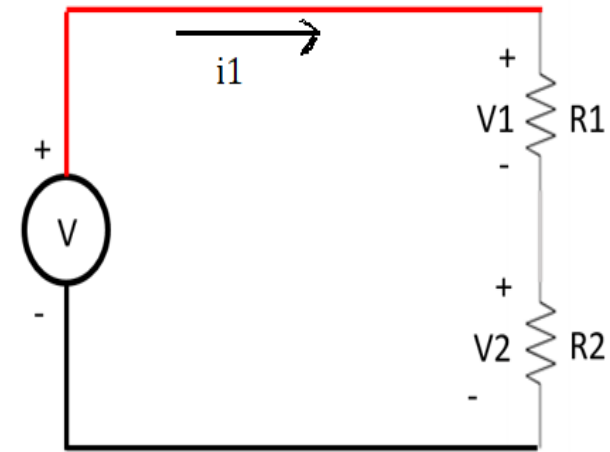
$$i_1 = \frac{V}{R_1 + R_2} \quad \text{Eq3}$$

- Now, the voltage across resistor R_1 is

$$V_1 = R_1 i_1 = \frac{R_1 V}{R_1 + R_2} \quad \text{Eq 4}$$

- Similarly

$$V_2 = R_2 i_1 = \frac{R_2 V}{R_1 + R_2} \quad \text{Eq 4}$$



Voltage Divider Rule Using more than two Resistor

- The connection of resistors in Figure 2 is said to be a series connection because all the elements carry the same current.
- To calculate the voltage across each resistance, we need to find the current in the loop ie. i_1
- To determine i_1 , we use KVL around the loop to obtain

$$V = V_1 + V_2 + V_3 + V_4 \quad \text{Eq1}$$

- Where, for example, V_1 is the voltage across the resistor R_1 . Using Ohm's law for each resistor,

$$V = R_1 i_1 + R_2 i_1 + R_3 i_1 + R_4 i_1 \quad \text{Eq2}$$

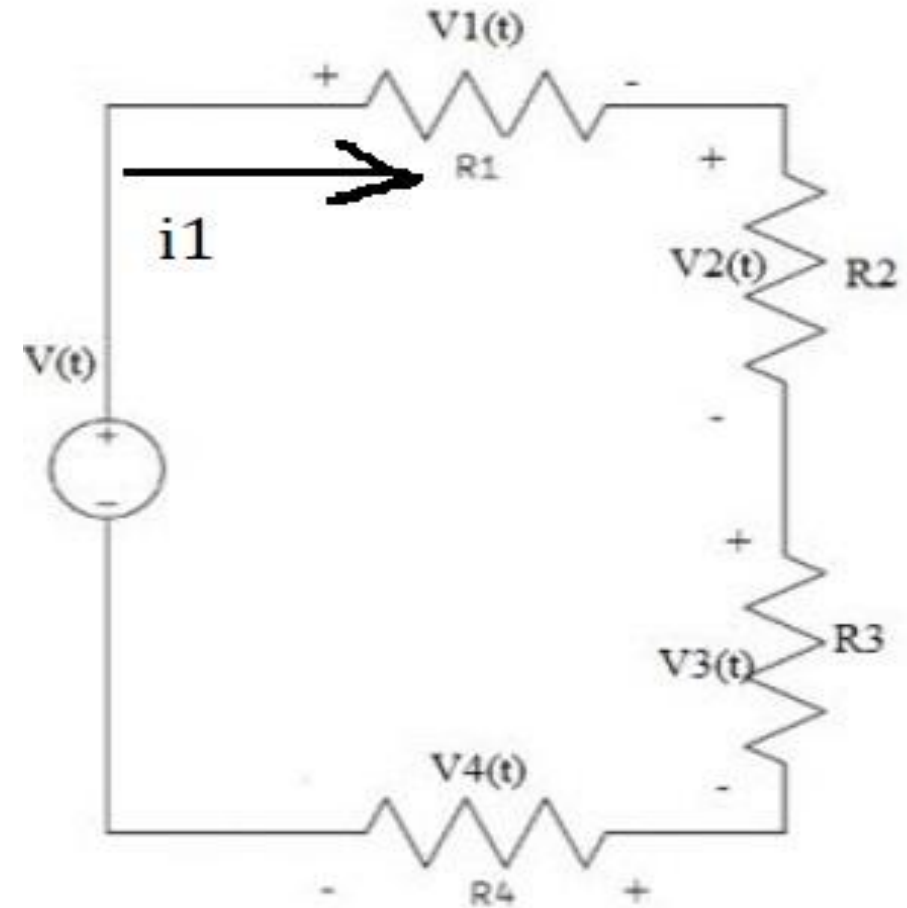


Figure 2

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- Solving for i_1 , we have

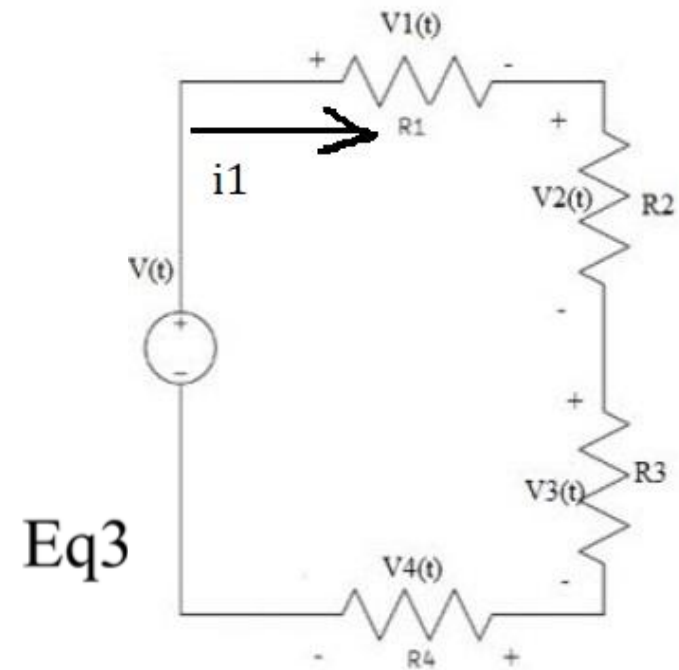
$$i_1 = \frac{V}{R_1 + R_2 + R_3 + R_4}$$

- Now, the voltage across resistor R_1 is

$$V_1 = R_1 i_1 = \frac{R_1 V}{R_1 + R_2 + R_3 + R_4} \quad \text{Eq 4}$$

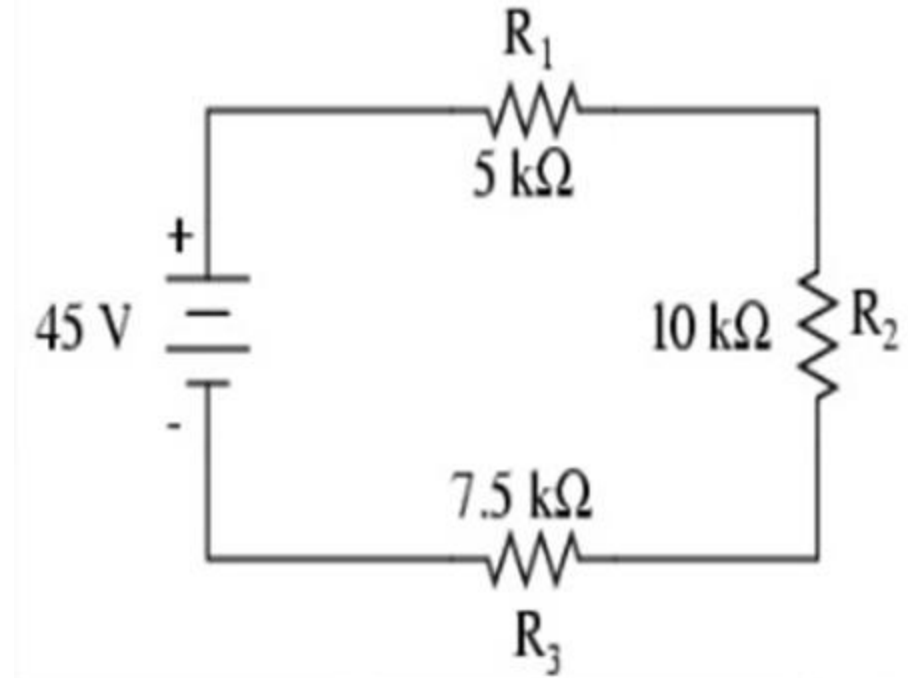
- Similarly, other voltages can also be determined
- Thus, the voltage across the n th resistor R_n is V_n and can be obtained as

$$V_n = \frac{R_n V}{R_1 + R_2 + \dots + R_n}$$



Example-1

Find the voltages across each resistors.



Example-1

Find the voltages across each resistors.

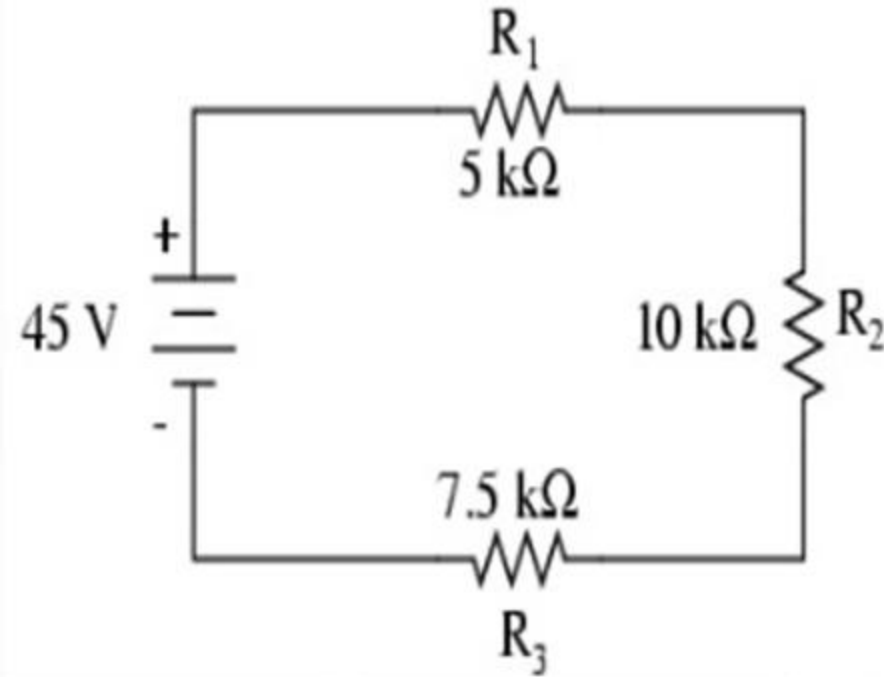
- $V_1 = (5/5+10+7.5) \times 45$
 $= 5/22.5 \times 45$

$$V_1 = 10V$$

- $V_2 = (10/5+10+7.5) \times 45$
 $= 10/22.5 \times 45$

$$V_2 = 20V$$

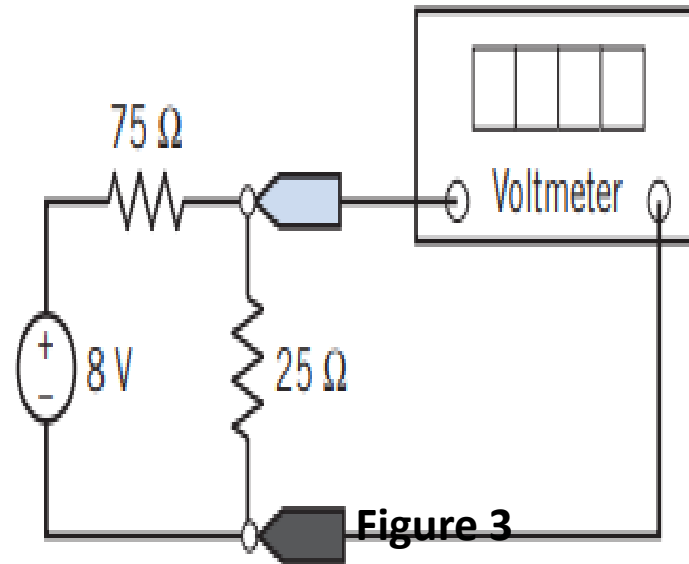
- $V_3 = (7.5/5+10+7.5) \times 45$
 $V_3 = 15V$



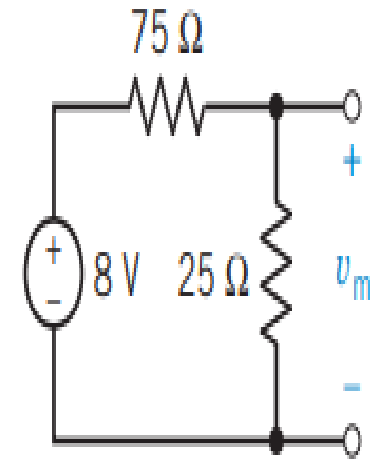
Practice Problem

- Determine the voltage measured by the voltmeter in the circuit shown in Figure 3

Ans $V_m = 2V$



(a)



(b)

Example 2

Determine the voltage across R_2 and R_4 using voltage division rule.

Assume

$V_1 = 20\text{ V}$, $R_1 = 10\Omega$, $R_2 = 5\Omega$, $R_3 = 30\Omega$ and $R_4 = 10\Omega$

Solution:

Directly voltage division rule cannot be applied, so first we need to find the equivalent resistance, ie. R_2 and R_4 are in series so,

$$\begin{aligned} R_{24} &= 5\Omega + 10\Omega \\ &= 15\Omega \end{aligned}$$

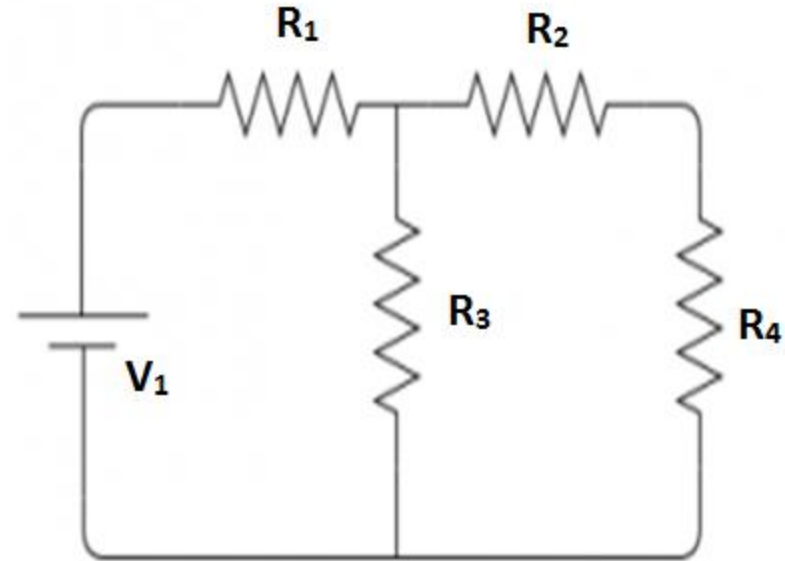


Figure 4

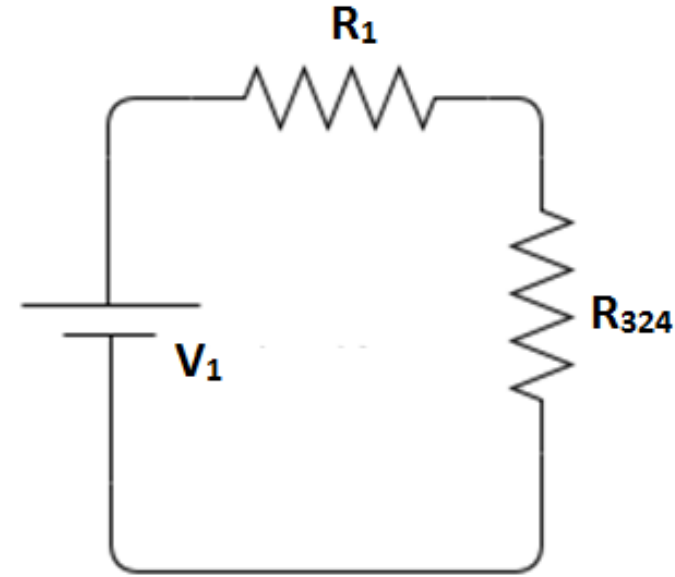
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- Then $R_{24} \parallel R_3$
 $R_{324} = 15\Omega \parallel 30\Omega$
 $= 10\Omega$

So, the circuit is simplified to the level shown in figure

And the voltage division rule can be applied directly:

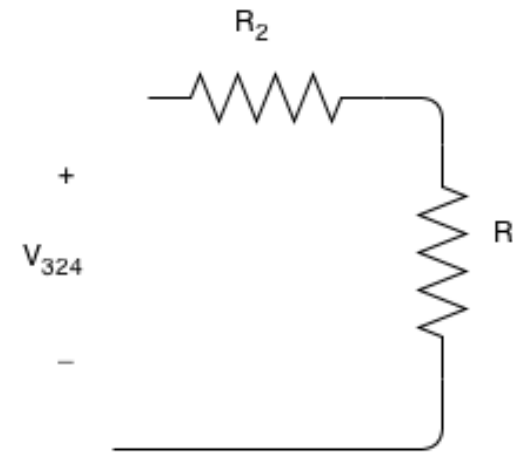
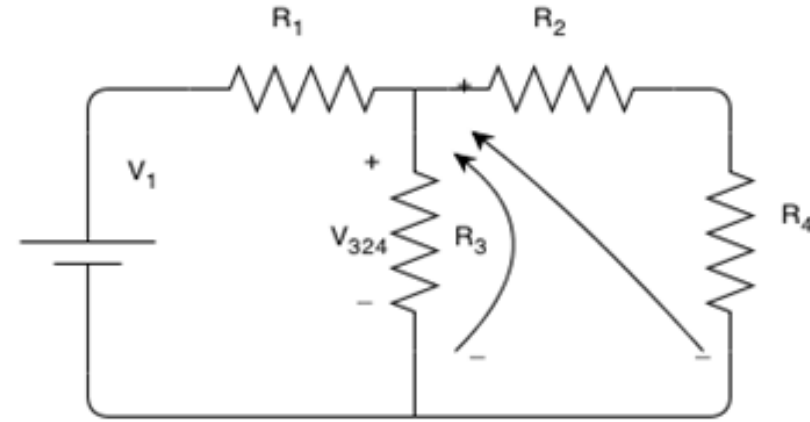
$$V_{R_{324}} = \frac{R_{324}}{R_{324} + R_1} V_1 = 10V$$



Contd..

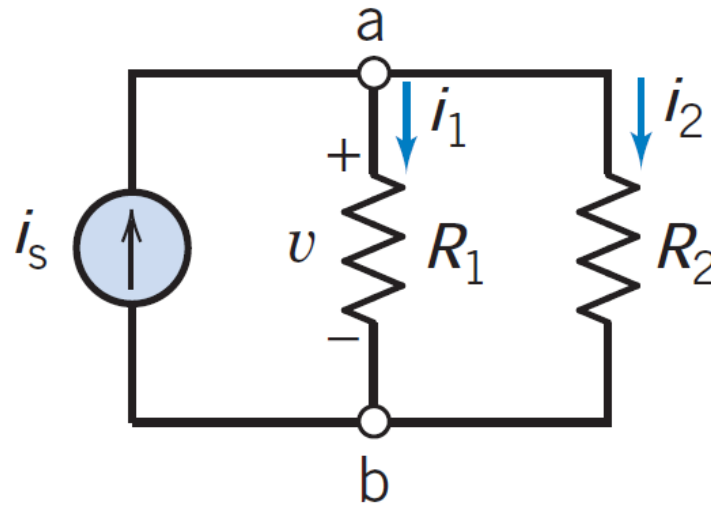
- Note that $V_{R_{324}}$ is the voltage across R_3 and the series combination of R_2 and R_4 as shown in Figure
- Now voltage division rule can be used to determine V_{R_2} and V_{R_4}
- Ans

$$V_{R_2} = \frac{10}{3} V$$
$$V_{R_4} = \frac{20}{3} V$$



Parallel resistors and Current division rule

- The circuit with two resistors and a current source shown in Figure.
- Note that both resistors are connected to terminals a and b and that the voltage v appears across each parallel element.

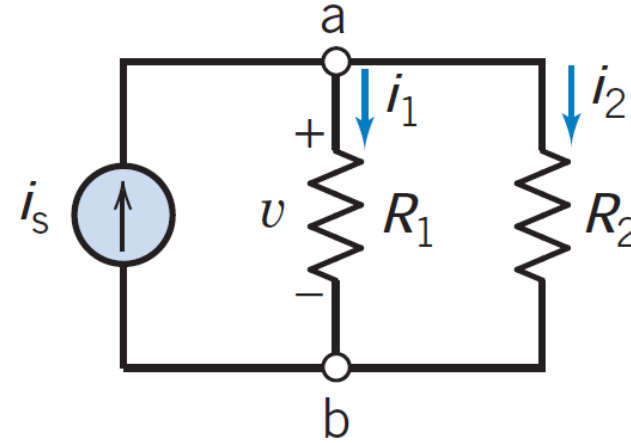


Parallel resistors and Current division rule

KCL at node a

$$i_s - i_1 - i_2 = 0$$

$$i_s = i_1 + i_2$$



From Ohm's law

$$i_1 = \frac{v}{R_1} \quad \text{and} \quad i_2 = \frac{v}{R_2}$$

$$i_s = \frac{v}{R_1} + \frac{v}{R_2}$$

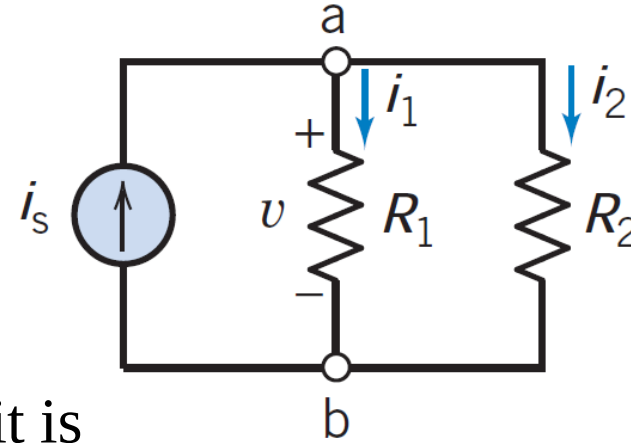
Parallel resistors and Current division rule

- Conductance G can be defined as the inverse of resistance R .

$$i_s = G_1 v + G_2 v = (G_1 + G_2) v$$

- Equivalent circuit for this parallel circuit is a conductance G_p , as shown in Figure ,

where $G_p = G_1 + G_2$



The equivalent resistance for the two-resistor circuit is found from

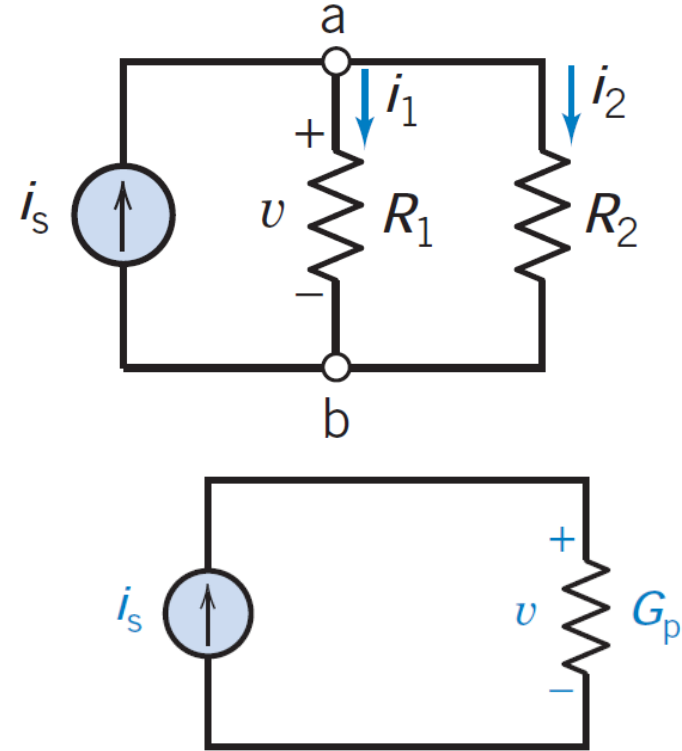
$$G_p = \frac{1}{R_1} + \frac{1}{R_2}$$

Parallel resistors and Current division rule

Because $G_p = 1/R_p$, we have

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_p = \frac{R_1 R_2}{R_1 + R_2}$$



- ❖ The circuit shown in Figure is **called a current divider circuit because it divides the source current.**

Parallel resistors and Current division rule

Note that $i_1 = G_1 v$

because $i_s = (G_1 + G_2)v$,

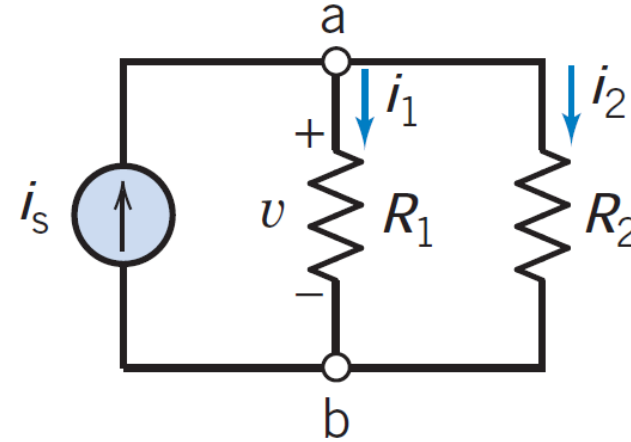
we solve for v ,
obtaining

$$v = \frac{i_s}{G_1 + G_2}$$

Substituting v , we obtain

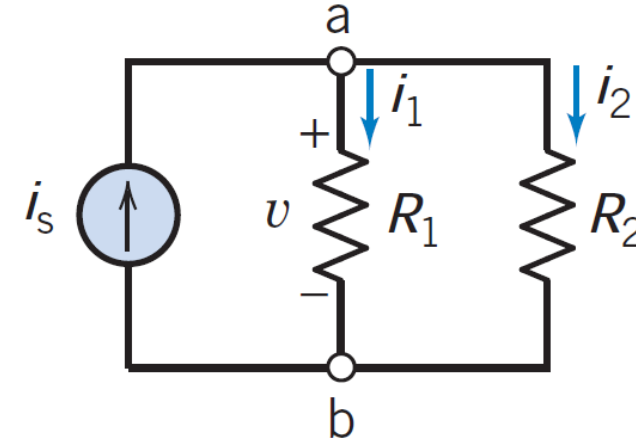
$$i_1 = \frac{G_1 i_s}{G_1 + G_2}$$

$$i_2 = \frac{G_2 i_s}{G_1 + G_2}$$



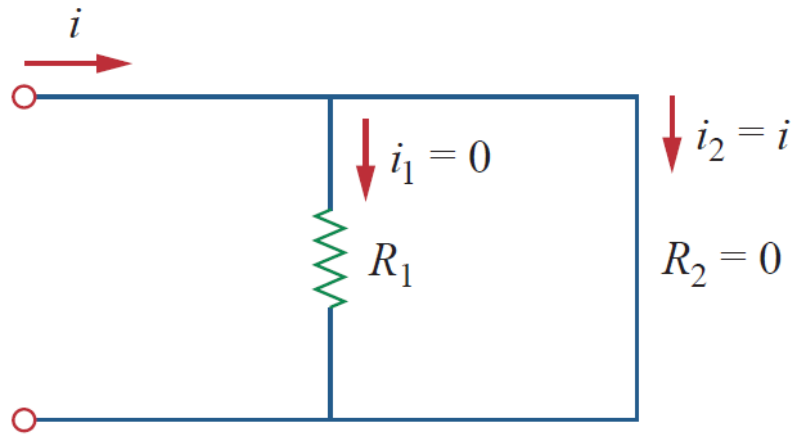
Parallel resistors and Current division rule

- Note that we may use $G_2 = 1/R_2$ and $G_1 = 1/R_1$ to obtain the current i_1 and i_2 in terms of two resistances as follows:



$$i_1 = \frac{R_2 i}{R_1 + R_2}, \quad i_2 = \frac{R_1 i}{R_1 + R_2}$$

- The current of the source divides between conductance G_1 and G_2 in proportion to their conductance values.

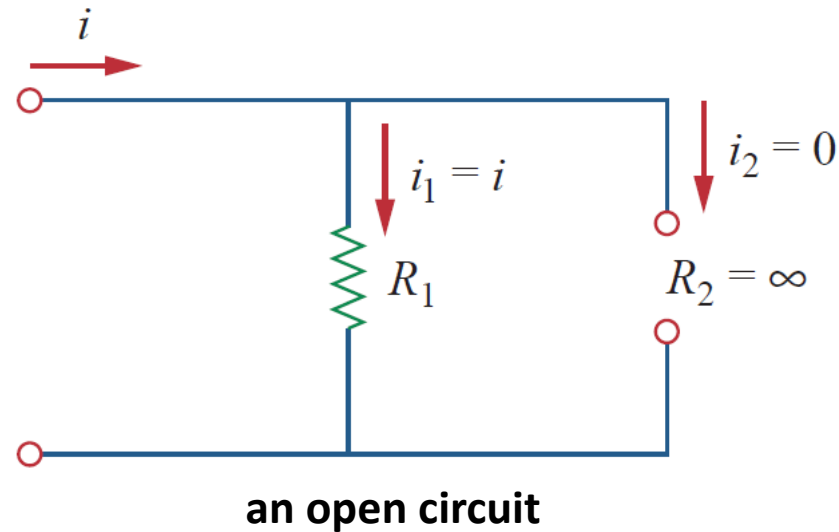


A shorted circuit

- In Fig. 1, suppose one of the resistors is zero, say $R_2 = 0$ or short circuit as shown in Fig. 3(a).
- From equation (10), $R_2 = 0$ implies that $i_1 = 0$, $i_2 = i$. This means that the current i bypasses R_1 and flows through the short circuit $R_2 = 0$, the path of least resistance.
- Thus, when a circuit is short circuited, the equivalent resistance,

$$R_{eq} = 0$$

and the entire current flows through the short circuit.



- Also, as an another extreme case, suppose $R_2 = \infty$ or R_2 is an open circuit.
- The current still flows through the path of least resistance, R_1 . By taking the limit of equation (7) as $R_2 \rightarrow \infty$, we obtain $R_{eq} = R_1$ in this case.

Parallel resistors and Current division rule

- Let us consider the more general case of current division with a set of N parallel conductors as shown in Figure.
- The KCL gives
$$i_s = i_1 + i_2 + i_3 + \cdots + i_N \quad \text{----- Eq. 1}$$

for which
$$i_n = G_n v \quad \text{for } n = 1, \dots, N$$

We may write Eq. 1 as
$$i_s = (G_1 + G_2 + G_3 + \cdots + G_N)v$$

Therefore,

$$i_s = v \sum_{n=1}^N G_n$$

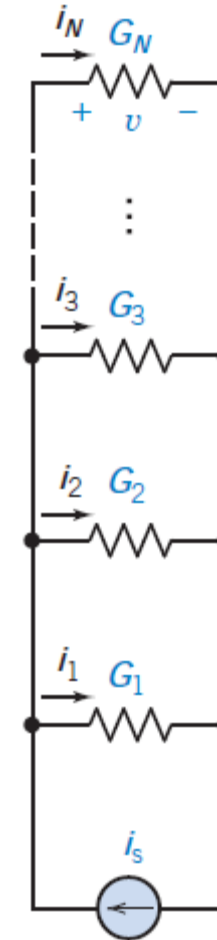


FIGURE Set of N parallel conductance with a current source i_s

Parallel resistors and Current division rule

Because $i_n = G_n v$

$$i_n = \frac{G_n i_s}{\sum_{n=1}^N G_n} \quad \text{we may obtain } v \text{ from} \quad i_s = v \sum_{n=1}^N G_n$$

Equivalent conductance G_p

$$G_p = \sum_{n=1}^N G_n$$

Therefore,
$$i_n = \frac{G_n i_s}{G_p}$$

which is the basic equation for the current divider with N conductances.

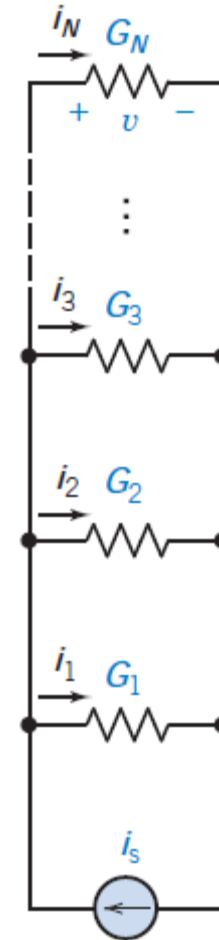
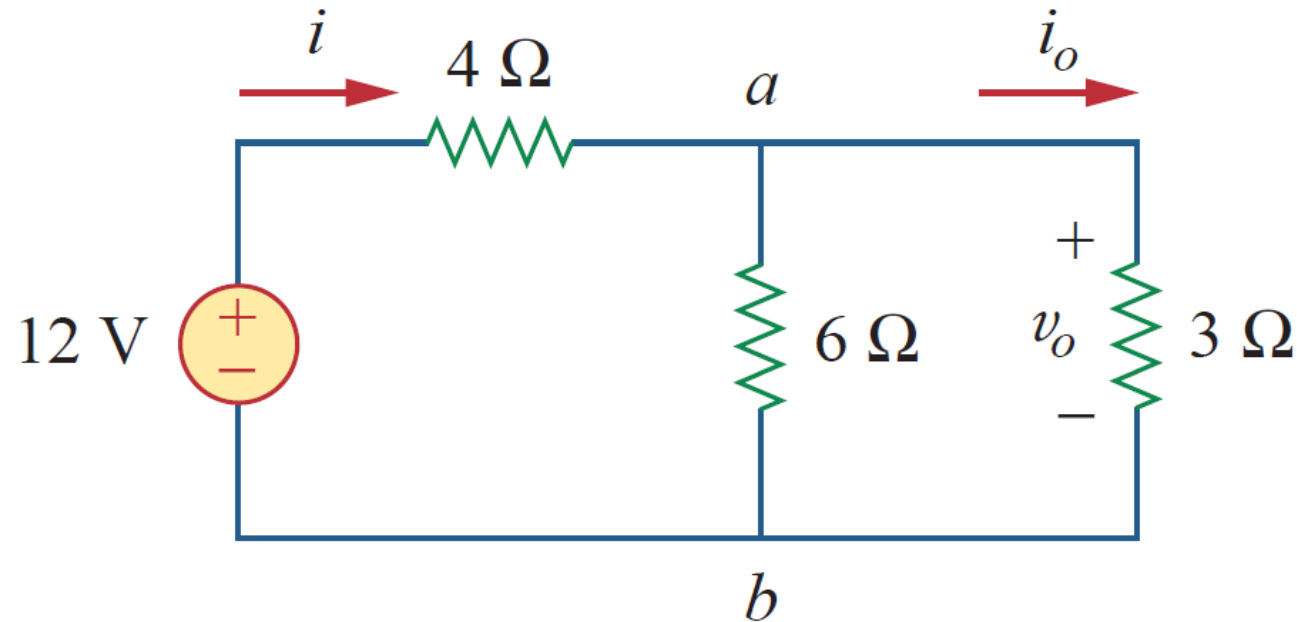


FIGURE Set of N parallel conductances with a current source i_s

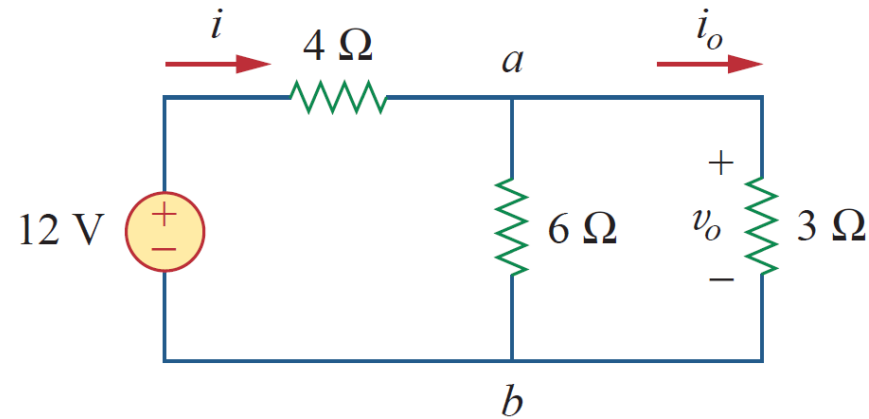
Problems

Problem-1

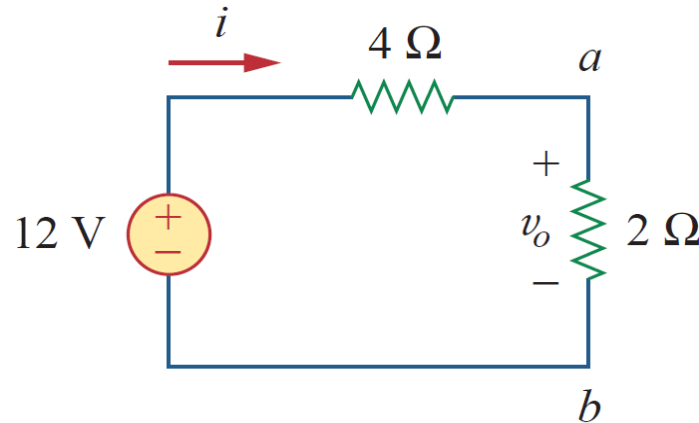
Find i_o and v_o in the circuit shown in Fig. Also, calculate the power dissipated in the $3\ \Omega$ resistor.



Solution:



$$6\ \Omega \parallel 3\ \Omega = \frac{6 \times 3}{6 + 3} = 2\ \Omega$$



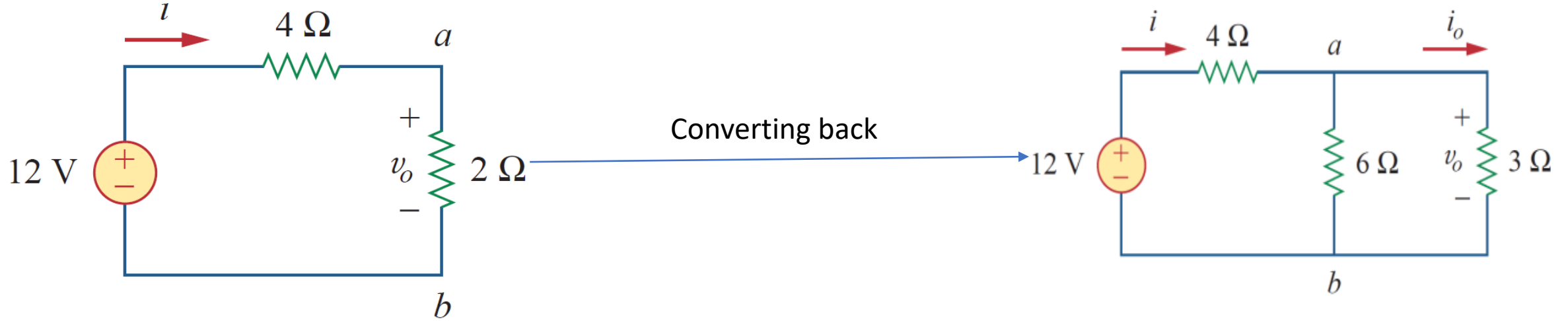
$$i = \frac{12}{4 + 2} = 2\ \text{A}$$

Voltage across the 6 Ω and 3 Ω resistors (i.e., across points a and b):

V_o = Voltage drop across 6 Ω or 3 Ω

$$= i \times R_p$$

$$= 2\ \text{A} \times 2\ \Omega = 4\ \text{V}$$



$$V_o = 3i_o$$

$$4 = 3i_o$$

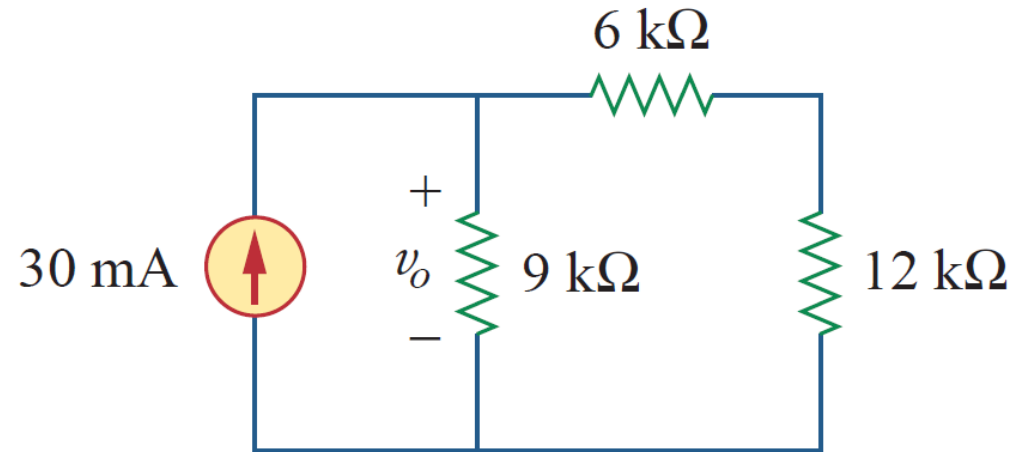
$$i_o = 4/3 \text{ A}$$

The power dissipated in the 3-Ω resistor is

$$p_o = v_o i_o = 4 \left(\frac{4}{3} \right) = 5.333 \text{ W}$$

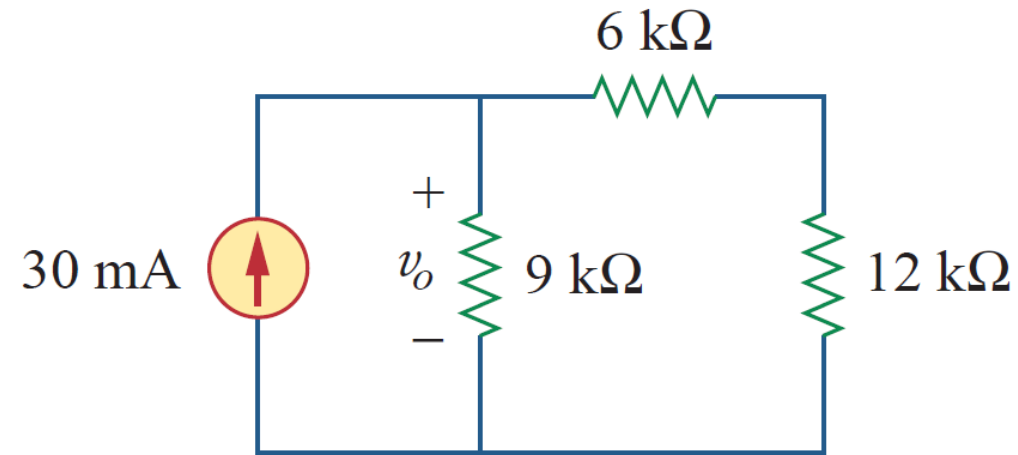
Problem-2

For the circuit shown in Fig., determine the (a) the voltage, v_o (b) the power supplied by the current source (c) the power absorbed by each resistor.



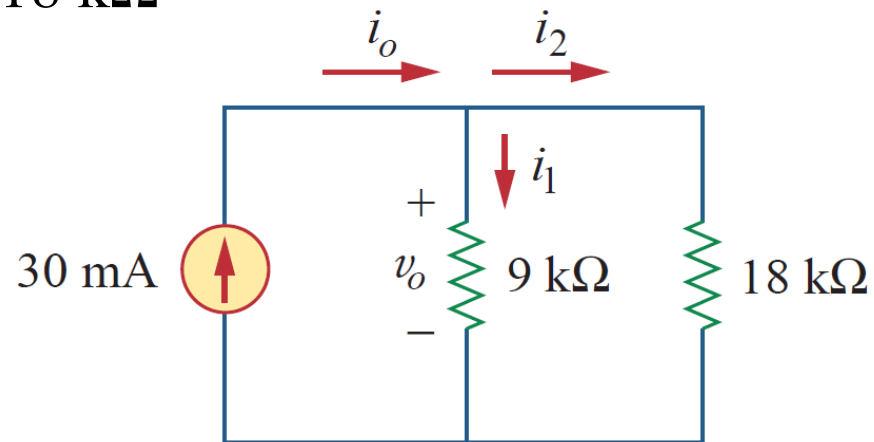
Solution:

(a)



6 kΩ and 12 kΩ resistors are in series,

So $6 + 12 = 18 \text{ k}\Omega$



We now apply the current division technique to find i_1 and i_2

$$i_1 = \frac{18,000}{9,000 + 18,000}(30 \text{ mA}) = 20 \text{ mA}$$

$$i_2 = \frac{9,000}{9,000 + 18,000}(30 \text{ mA}) = 10 \text{ mA}$$

Notice that the voltage across the 9k Ω and 18k Ω resistor is the same.

$$V_0 = i_0 * R_p = 30\text{mA} * 6\text{k } \Omega = 180\text{V}$$

$$\text{Or } V_0 = i_1 * R_1 = 20\text{mA} * 9\text{k}\Omega = 180\text{V}$$

$$\text{Or } V_0 = i_2 * R_2 = 10\text{mA} * 18\text{k}\Omega = 180\text{V as expected}$$

(b) Power supplied by the source is

$$p_o = v_o i_o = 180(30) \text{ mW} = 5.4 \text{ W}$$

(c) Power absorbed by the 12-k Ω resistor is

$$p = iv = i_2(i_2R) = i_2^2R = (10 \times 10^{-3})^2 (12,000) = 1.2 \text{ W}$$

Power absorbed by the 6-k Ω resistor is

$$p = i_2^2 R = (10 \times 10^{-3})^2 (6,000) = 0.6 \text{ W}$$

Power absorbed by the 9-k Ω resistor is

$$p = \frac{v_o^2}{R} = \frac{(180)^2}{9,000} = 3.6 \text{ W}$$

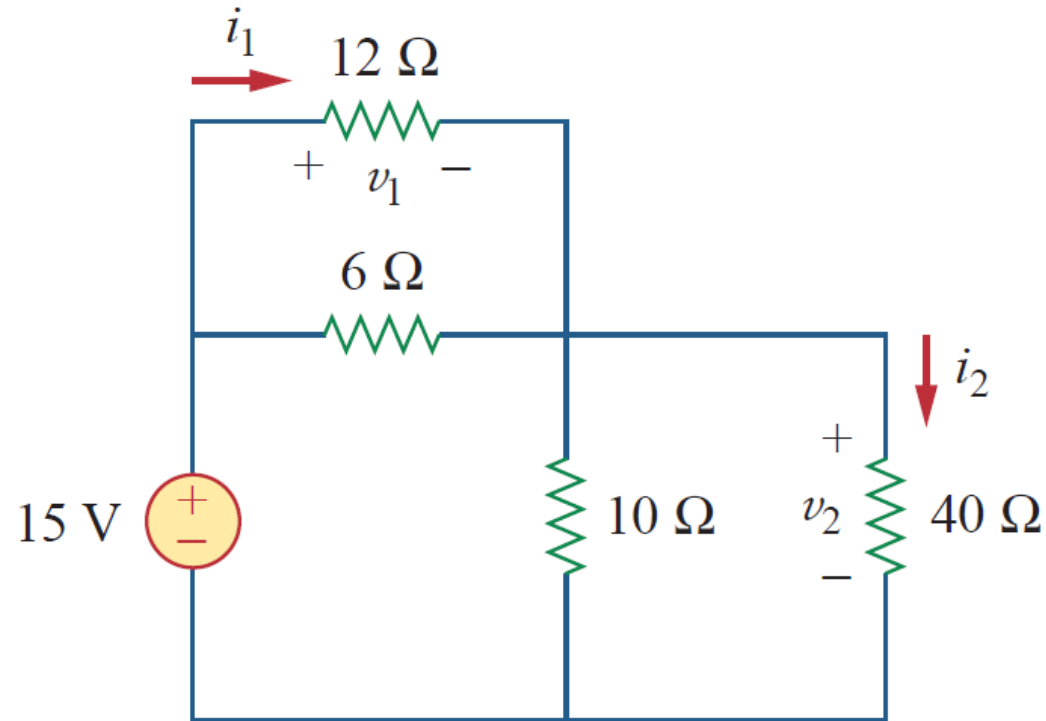
or

$$p = v_o i_1 = 180(20) \text{ mW} = 3.6 \text{ W}$$

Notice that the power supplied (5.4 W) equals the power absorbed ($1.2 + 0.6 + 3.6 = 5.4 \text{ W}$). This is one way of checking results.

Problems

Find v_1 and v_2 in the circuit shown in Fig. Also, calculate i_1 and i_2 and the power dissipated in the $12\ \Omega$ and $40\ \Omega$ resistors.

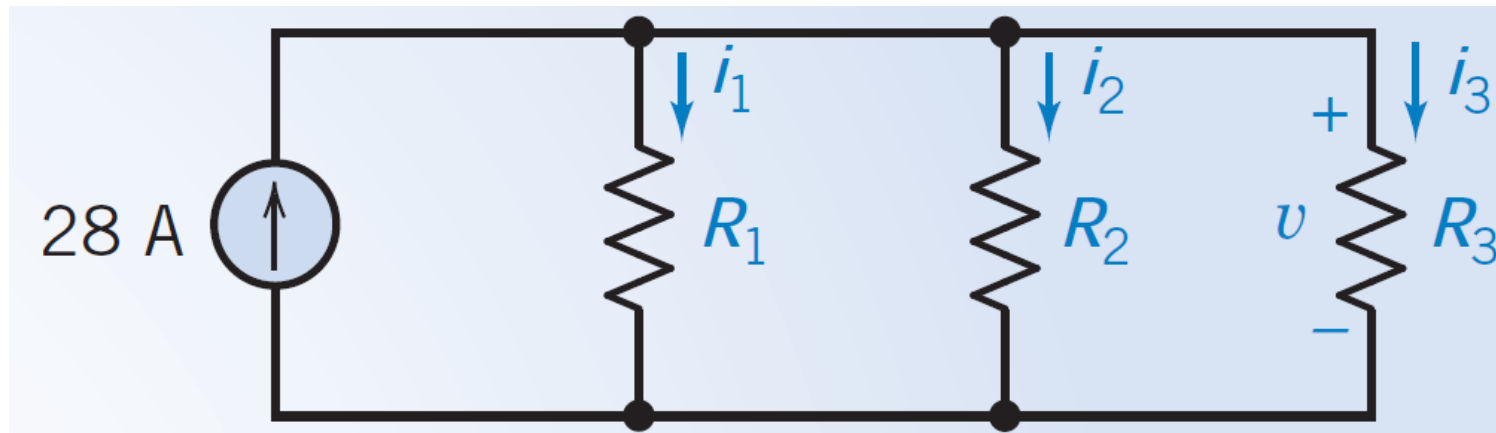


Answer: $v_1 = 5\text{ V}$, $i_1 = 416.7\text{ mA}$, $p_1 = 2.083\text{ W}$, $v_2 = 10\text{ V}$, $i_2 = 250\text{ mA}$, $p_2 = 2.5\text{ W}$.

Practice Problems

For the circuit in Figure , find (a) the current in each branch, (b) the equivalent circuit, and (c) the voltage v .

The resistors are $R_1 = 1/2 \, \Omega$; $R_2 = 1/4 \, \Omega$; $R_3 = 1/8 \, \Omega$



Ans: Current $i_1 = 4 \text{ A}$, $i_2 = 8 \text{ A}$, $i_3 = 16 \text{ A}$

References

1. R.C. Dorf and James A. Svoboda, “Introduction to Electric Circuits”, 9th ed, John Wiley & Sons, 2013.
2. D.C. Kulshreshtha, Basic Electrical Engineering, Revised 1st ed, Tata Mc Graw Hill, 2017.
3. V. K.Mehta, Rohit Mehta, Basic Electrical Engineering, 6th ed, S. Chand Publishing, 2012.